

Restricted attention over an intact, position-preserving KV cache

A Retain the prefill



B Build a per-query mask

one decode row of additive mask $M[b,h,q,k]$

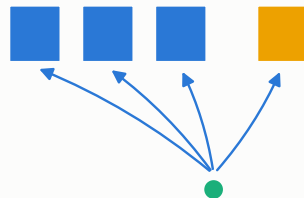


blue = selected or always-keep

gray = hidden for this query only

C Decode in place

attend only to allowed keys



next token

No gather · no eviction · no re-read